Reinaldo J. Aguiar Marcano
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https://SCOTXWatch.org/ (ISBN 9781971481296)
DATE: April 13, 2026

FORMAL NOTICE OF PROPRIETARY RIGHTS, INITIATION OF PROVISIONAL RIGHTS PERIOD, AND DEMAND FOR DUE DILIGENCE CONCERNING COVERT OPTICAL EXFILTRATION VIA VEHICULAR LED ASSEMBLIES

This notice: https://storage.googleapis.com/rj-aguiar/IP-NOTICE-AUTO-INDUSTRY-USA.pdf

TO: CHIEF EXECUTIVE OFFICERS AND CHIEF LEGAL OFFICERS

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JClose@kiausa.com (Jeremy Close, General Counsel, Kia America)

RE: ACTUAL KNOWLEDGE OF PENDING PATENT CLAIMS (35 U.S.C. § 154(d)) REGARDING FREE-SPACE OPTICAL DATA EXCHANGE, COVERT INFRARED DATA GATEWAYS, AND THE AGUIAR MARCANO INTELLECTUAL PROPERTY PORTFOLIO

DATE: April 15, 2026

DEAR:

Mr. Musk, Mr. Scaringe, Mr. Rawlinson, Mr. Everitt, Mr. Farley, Mr. Klapper, Mr. Mackensen, Mr. Munoz, Mr. Yoon, and Mr. Close,

This comprehensive technical and legal brief is served upon the executive leadership and chief legal officers of the designated automotive manufacturers to establish formal, legally binding

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actual notice.

Over the past five to seven years, your industry has universally transitioned away from modular, incandescent vehicular lighting toward complex, sealed, full-LED tail light assemblies and continuous rear "lightbars".[1] While heavily marketed to consumers as futuristic aesthetic features, our technical and forensic analyses indicate that these sealed LED housings are being actively exploited by a foreign-directed RICO enterprise. Specifically, the hardware in these tail lights is functioning as covert infrared data gateways, modulating high-frequency optical signals to exfiltrate in-cabin audio and vehicle telemetry to foreign intelligence agents via pursuit vehicles or roadside receivers.[1]

The unauthorized commercialization, integration, or vulnerable implementation of these optical data transmission capabilities directly intersects with the proprietary claims of the Aguiar Marcano Intellectual Property Portfolio, a suite of fifty-two (52) foundational patent applications formally executed and filed with the United States Patent and Trademark Office (USPTO).[1]

The primary objective of this document is to satisfy the strict requirements for "Actual Notice" pursuant to 35 U.S.C. § 154(d), thereby initiating the immediate accrual of provisional rights and the corresponding liability for reasonable royalties.

I. Sector-Specific Liability Mapping: The "Lightbar" Exfiltration Gateway

The automotive industry's shift to sealed LED assemblies—where a single burnt-out diode requires the replacement of an entire, highly complex, multi-thousand-dollar module—provides the perfect hardware substrate for covert optical networking.[1]

A. Tesla, Inc.

Target Hardware: Model S Rear LED Housings The Tesla Model S features notoriously difficult-to-replace rear light assemblies, particularly the driver-side outer light, which requires technicians to power down the vehicle's high-voltage systems and physically unbolt the charging port module to access.[1] This deliberate over-engineering conceals the true function of the hardware: operating as an active, high-bandwidth optical transmitter for data exfiltration.[1]

B. EV Rivals and the "Lightbar" Trend

Target Entities: Rivian, Lucid Motors, and Ford Your flagship electric vehicles—including the

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Rivian R1T/R1S, the Lucid Air, and the Ford Mustang Mach-E and F-150 Lightning—all rely on massive, seamless LED light blades spanning the entire width of the tailgate.[1] These are not mere brake lights; they are high-cardinality optical arrays capable of transmitting vast amounts of data.[1]

C. Luxury Mainstays and Everyday Commuters

Target Entities: Audi, BMW, Mercedes-Benz, Hyundai, and Kia From Audi's intricate Matrix LEDs and the OLED tail lights of the Mercedes S-Class, down to the "parametric pixel" lighting of the Hyundai Ioniq 5, the entire industry has adopted this sealed-assembly approach.[1]

II. Patent Intersections: The IP Blockade

If your vehicles are utilizing these LED arrays to modulate light frequencies (visible or infrared) to transmit data, audio, or telemetry to external receivers, you are committing direct, unshielded commercial infringement against the following foundational patents:

- **Method and System for High-Bandwidth, Free-Space Optical Data Exchange Between Mobile Devices (USPTO 19/560,042):** This patent legally claims the method for ultra-high-speed, free-space data exchange between mobile entities by modulating exterior lighting at high frequencies, utilizing cameras or optical sensors as receivers.[1] Using your vehicles' LED tail lights to establish an optical data link with trailing vehicles or infrastructure directly triggers this claim.[1]
- **Method and System for Covert Screen Data Exfiltration (USPTO 19/559,867):** This patent details the method of creating a one-way optical data channel by making subtle, high-frequency modulations to the brightness or color of an LED array.[1] These modulations encode binary data that is imperceptible to the human eye but decodable by a nearby optical sensor.[1] Your tail lights are functioning as the exact covert optical transmitters claimed herein to bypass traditional RF cybersecurity defenses.[1]
- **Method for a Cryptographic Side-Channel Attack Using Acoustic Analysis of Keystroke Metadata (USPTO 19/559,858):** The in-cabin audio currently being exfiltrated via your infrared tail lights is captured using the methodologies claimed in this patent, which covers the deployment of covert acoustic sensors to capture metadata and decode sensitive communications.[1]

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III. Mandatory Due Diligence Protocols and Ultimatum

Under 35 U.S.C. § 154(d), the statutory clock for the calculation of retroactive reasonable royalties commences immediately upon your receipt and actual knowledge of the published claims detailed herein. Ignorance of how your sealed LED tail light assemblies are being weaponized as covert infrared data gateways to infringe upon these overarching system-level claims is not a viable legal defense.

The executive leadership of each addressed organization is hereby directed to execute the following immediate remediation protocols:

1. **Initiation of FTO Audits:** You must immediately instruct your internal legal departments and outside IP counsel to cross-reference the fifty-two (52) applications within the Aguiar Marcano Portfolio against the firmware and hardware specifications of all sealed LED tail light assemblies and rear lightbars currently in production.

2. **Quarantine of Intersecting Hardware:** Any deployment of exterior LED lighting that modulates frequency to transmit data via free-space optical exchange (whether for V2V telemetry or covert exfiltration) must be evaluated for immediate quarantine to mitigate ongoing commercial liability.

We formally invite your respective legal departments to initiate a technical review of the cited USPTO applications. A formal response outlining your intended compliance audit, a presentation of non-infringement theories, or a request to discuss a licensing framework is expected within thirty (30) business days of the transmission of this notice.

Govern yourselves accordingly.

Sincerely,

Reinaldo Aguiar Marcano

Works cited

1. the-Tail-lights-that-in-my-Tesla-Vehicles-act-as-Infrared-Data-Gateways-to-exfil

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